Limits to Arbitrage in the Indian Equity Market

Stage I: theory, data and the futures-cash band

A BS project report (Stage I, EC 495) submitted in partial fulfilment of the requirements for the award of the degree of

Bachelor of Science in Economics

by

Prince Kumar
(Roll No. 23B3032)

Under the supervision of

Prof. Aditi Chaubal



Department of Economics
INDIAN INSTITUTE OF TECHNOLOGY BOMBAY
Mumbai 400076, India

November 2026

Certificate

This is to certify that the report entitled **Limits to Arbitrage in the Indian Equity Market, Stage I**, submitted by **Prince Kumar** (Roll No. 23B3032) to the Indian Institute of Technology Bombay as part of the requirements for the degree of **Bachelor of Science in Economics**, is a record of original, bona fide research work carried out under my supervision and guidance.

The results contained in this report have not been submitted in part or in full to any other university or institute for the award of any degree or diploma, to the best of my knowledge.

Prof. Aditi Chaubal
Department of Economics
Indian Institute of Technology Bombay

Declaration

I declare that this written submission represents my ideas in my own words and that, where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented, fabricated or falsified any idea, data, fact or source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have not been properly cited or from whom proper permission has not been taken when needed.

Prince Kumar
Roll No. 23B3032
Date:
Place: IIT Bombay

Acknowledgements

I thank my supervisor, Prof. Aditi Chaubal, for her guidance throughout this project, and in particular for the advice to build a formal model of what the project would test before looking at any data. That advice shaped the whole design of the work.

Prince Kumar

Contents

Certificate i

Declaration ii

Acknowledgements iii

Abstract 1

1 Introduction 1

2 The Indian setting 4

2.1 Stock futures . . . 4
2.2 Short selling . . . 4
2.3 The position limit and the ban . . . 5
2.4 Trading costs and arbitrage capital . . . 5

3 Related literature 6

3.1 Why arbitrage is limited . . . 6
3.2 The no-arbitrage band . . . 7
3.3 Evidence from India . . . 7

4 A model of arbitrage under India's limits 7

4.1 Set-up . . . 8
4.2 The no-arbitrage band . . . 9
4.3 Existing positions . . . 10
4.4 The F&O ban . . . 10
4.5 From the model to the tests . . . 11

5 Research design and data 12

5.1 How the work was organised . . . 12
5.2 Data . . . 13
5.3 What the data show . . . 15

6 The no-arbitrage band 17

6.1 A threshold model . . . 17
6.2 Estimates . . . 17
6.3 How much of the reversion is noise . . . 17
6.4 Smooth adjustment . . . 19
6.5 The band over time and over the life of a contract . . . 19
6.6 Bands of individual stocks . . . 20

7 What sets the band 20

7.1 Tax increases (pre-specified test H3) . . . 20
7.2 What an arbitrage position costs . . . 21
7.3 Frictions across stocks (H1) . . . 22

8 Does the ban bind on trading? 23

8.1 Trading on ban days (pre-specified test H8) . . . 23
8.2 Does every trade close a position? . . . 24
8.3 At the 95% trigger . . . 25
8.4 How the 2025 rules changed the number of bans . . . 26

9 Conclusion and the second stage 28

9.1 What this stage shows . . . 28
9.2 Limitations . . . 29
9.3 The second stage . . . 29

References 30

A Proofs 31

B The test plan and the data **32**
B.1 The pre-analysis plan 32
B.2 Data construction 32

BS PROJECT, STAGE I (EC 495) | DEPARTMENT OF ECONOMICS, IIT BOMBAY | NOVEMBER 2026

Limits to Arbitrage in the Indian Equity Market: Theory, Data and the Futures-Cash Band

Prince Kumar Supervisor: Prof. Aditi Chaubal

ABSTRACT

A stock future should trade at the price of the stock plus the cost of carrying it to expiry; otherwise an arbitrageur could buy the cheaper of the two and sell the dearer. Trading costs, short-sale constraints, limited capital and position limits make this arbitrage costly, so the gap between the two, the basis, can persist. In this first stage of my BS project I study the basis of single-stock futures in India from 2015 to 2026. I build a model in which each Indian limit to arbitrage is a parameter, including the F&O ban, under which every trader may only reduce positions once open interest nears a stock's position limit, and I fix the tests before relating any outcome to any treatment. The basis moves in a wide, asymmetric band, from about 35 basis points below each stock's median to about 24 above it, and the full cost of a new cash-and-carry position lies in its outer tail. The upper part of the basis rose after the 2024 and 2026 tax increases, as predicted, but by more than the cost changes and by amounts that are not unusual on other dates. The ban binds hard on trading: futures volume falls by about 87%, and under intraday enforcement every futures trade in a ban closed existing positions.

Keywords: *limits to arbitrage; futures basis; no-arbitrage band; threshold autoregression; position limits; India*

JEL codes: *G13, G14, G18*

BAND AROUND THE MEDIAN	COST OF A NEW POSITION	FUTURES VOLUME IN A BAN
−35 to +24 bp	27 to 31 bp	−87%
BAN SPELLS, 2015 TO 2026	**BASIS OBSERVATIONS**	**BAN DAYS, 2025 RULE VS OLD**
1,788	486,040	−73%

Band: pooled threshold model, basis relative to each stock's median. Cost: round trip of an overnight cash-and-carry position held to delivery. Volume: effect of a ban on futures contracts traded. Ban days: first year of the October 2025 rules against the old rule applied to the same stocks.

1. INTRODUCTION

There are two ways to hold a share of a company for the next month. One is to buy the share today. The other is to buy a futures contract that delivers the share at the end of the month. Since both end in the same place, their prices should be linked: the futures price should equal

today's share price, plus the interest one would pay to finance the share, minus any dividend paid before delivery. If the future is dearer than this *cost-of-carry* value, a trader can buy the share, sell the future and hold both until delivery, earning the difference without any exposure to the share price. This is *cash-and-carry arbitrage*. If the future is cheaper, the reverse trade (sell the share, buy the future) earns the difference. The gap between the futures price and its cost-of-carry value is called the *basis*. In a market where these trades are free and unlimited, the basis should be close to zero at all times.

In real markets arbitrage has limits. Every trade pays taxes and fees. The reverse trade needs a short sale, which may be costly or not allowed. An arbitrage position ties up capital, and the gap can widen before it closes, so the arbitrageur bears risk and may be forced to close early (Shleifer and Vishny, 1997; Gromb and Vayanos, 2010). Exchanges also cap the positions traders may hold. With such limits, the basis can wander inside a band without anyone finding it worth closing, and arbitrage pushes back only when the basis leaves the band. How wide this band is, and which limits set its edges, are empirical questions.

India is an unusually good place to answer them, because most of its limits to arbitrage have changed on known dates. The securities transaction tax (STT) on the sale of futures was raised in April 2023, October 2024 and April 2026. Only stocks with derivatives, the stocks of the futures and options (F&O) segment, may be sold short, and in practice the overnight short position in an Indian stock is a short future. Arbitrage mutual funds, which hold exactly the cash-and-carry trade, grew about fourfold after a change in the taxation of debt funds in 2023. And position limits are set by formula and were redesigned in October 2025.

India also has a rule that, to my knowledge, has not been studied. Each stock's derivatives have a *market-wide position limit* (MWPL). When open interest, the number of contracts outstanding across all traders, exceeds 95% of this limit at the end of a day, the stock enters an *F&O ban* the next day. During the ban every trader, not only arbitrageurs, may only trade to reduce positions, and the ban lasts until open interest falls to 80% of the limit. Between 2015 and October 2026 this happened 1,788 times, to 141 different stocks. A ban suddenly removes the possibility of opening new arbitrage positions in a stock, which makes it a sharp test of what position limits do.

The project asks how strongly these limits bind in Indian equities and how much mispricing they explain. It has two stages. This report covers the first. It sets out a model of arbitrage under India's limits, fixes the tests in advance, builds the data, and answers three questions: how wide is the band within which the basis of Indian stock futures moves; do trading costs and other frictions set its edges; and does the F&O ban actually restrict trading in the way the rule intends? The second stage asks what the ban and the other limits do to prices.

I proceed in a fixed order, designed to keep theory and data from being fitted to each other, which was the main advice of my supervisor at the start of the project. I first wrote a model in which each limit enters as a parameter. From the model I derived predictions, each with the result that would contradict it. I then wrote these tests into a pre-analysis plan and froze it with a time stamp before I related any outcome to any treatment. The data, built from public files of the National Stock Exchange (NSE), cover 486,040 stock-days of the futures-cash basis for 371 stocks.

The findings of this stage are the following.

1. **The basis moves in a wide, asymmetric band.** A three-regime threshold model places the edges of the band about 35 basis points (bp) below and 24 bp above each stock's typical basis. Inside the band the basis drifts; outside it, it moves back. The lower edge is further away, as expected when the reverse trade, which needs a short sale, is costlier. A test of the linear model against the threshold model rejects linearity decisively. Part of the day-to-day reversion that the model measures is, however, noise in daily prices rather than arbitrage.
2. **Costs shape the band, but loosely.** The full cost of opening a new cash-and-carry position and holding it to delivery is 27 to 31 bp, which places it in the outer tail of the basis, while rolling an existing position costs only 4 to 8 bp. The upper part of the basis distribution rose after the 2024 and 2026 tax increases, as the model predicts, and the test fixed in advance supports this. But the rise was larger than the change in costs and was not unusual compared with the same statistic on other dates, so I read the evidence as weak. Across stocks, the band is wider where trading frictions are higher.
3. **The ban binds hard on trading.** On ban days futures trading falls by about 87% relative to the same stock's days at similar levels of open interest, and cash-market trading moves toward delivery. How the rule was enforced matters: once violations were judged on intraday rather than end-of-day positions, from March 2020, every futures trade during a ban closed an existing position. There is no sign that traders hold back to avoid the trigger.
4. **The 2025 rules cut the number of bans.** Measured against the old rule applied to the same stocks, the October 2025 rules produced about a quarter of the ban days in their first year.

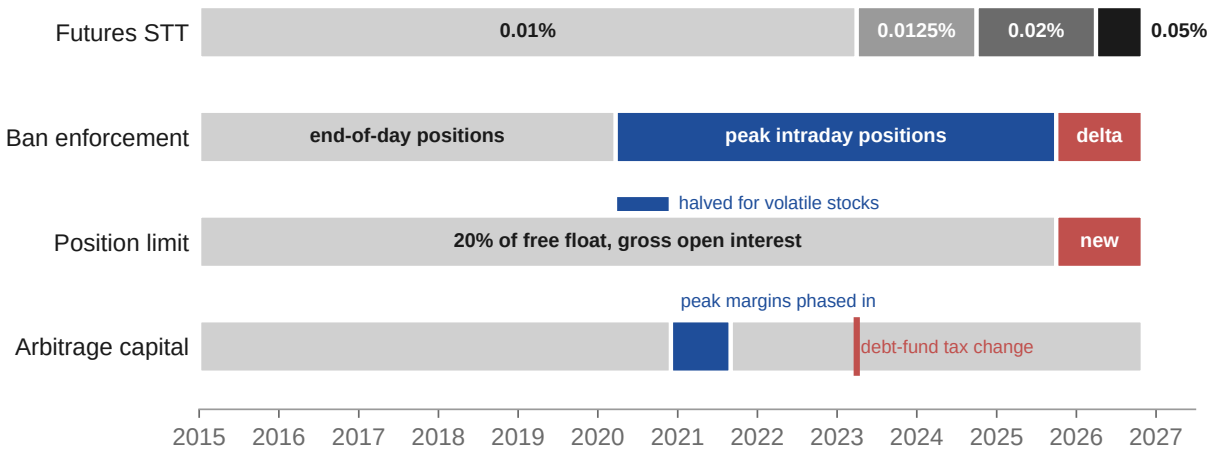
The closest earlier studies of Indian stock futures estimate no-arbitrage bands from intraday data over a few months or years (Aggarwal, 2015; Shankar, Sankar and Kiran Kumar, 2019). This report adds a model of arbitrage that includes India's position limits and the ban, estimates the band over twelve years of daily data for every stock with futures, relates it to dated changes in costs, and gives the first evidence on how the F&O ban affects trading.

Section 2 describes the Indian rules. Section 3 reviews the related literature. Section 4 presents the model and its predictions. Section 5 describes the research design and the data. Section 6 estimates the band, Section 7 asks what sets it, and Section 8 studies trading in a ban. Section 9 concludes and sets out the second stage. The appendices give the proofs and the details of the test plan and the data.

FIGURE 1

Almost every limit to arbitrage changed on a known date during the sample

Futures STT, enforcement of the ban, the position-limit rule and arbitrage capital, 2015 to 2026



Source: SEBI circulars and press releases; NSE circulars (listed in the references).

2. THE INDIAN SETTING

This section describes the rules that shape each limit to arbitrage in Indian stock futures. Figure 1 places the dated changes on one time line.

2.1 Stock futures

Single-stock futures trade on the NSE for about two hundred large and liquid stocks, the members of the F&O segment. A stock enters the segment when it meets eligibility criteria set by the Securities and Exchange Board of India (SEBI), based on market capitalisation, traded value and delivery, and leaves it when it fails them. Contracts expire monthly, three months are listed at a time, and most trading is in the nearest month. Since October 2019 stock derivatives are settled by delivery of the shares. One detail of the exchange's prices matters for measurement: when a future does not trade in the last half hour of a day, the exchange sets its settlement price at a theoretical cost-of-carry value. A basis computed from settlement prices would therefore look small on thin days for purely mechanical reasons, so I use prices from actual trades throughout.

2.2 Short selling

SEBI's framework allows all investors to sell short, but only in securities of the F&O segment, and it forbids naked short selling: every short sale must be delivered at settlement, so an overnight short needs borrowed shares. Shares can be borrowed through the exchange's securities lending and borrowing (SLB) segment, but it is thin: on a typical day only a few dozen stocks are lent. In practice the overnight short position in an Indian stock is a short future. This is why the reverse arbitrage trade (short the stock, buy the future) is much costlier than cash-and-carry, and why

TABLE 1

The limit was cut and open interest redefined in October 2025

Market-wide position limit (MWPL) and the open interest tested against it

PERIOD	MWPL	OPEN INTEREST TESTED
To 30 Sep 2025	20% of shares held by non-promoters (free float)	Gross open interest, futures and options
23 Mar to 26 Nov 2020	Halved for volatile or heavily used stocks	Gross open interest
From 1 Oct 2025	Lower of 15% of free float and 65 times average daily delivery (in shares), floor 10% of free float; recomputed quarterly	Futures-equivalent (delta-adjusted) open interest

Source: SEBI circulars of 22 September 2006 and 29 May 2025; SEBI press release 18/2020; NSE Clearing circular 119/2025.

F&O membership defines which stocks can be shorted at all.

2.3 The position limit and the ban

The MWPL caps the total open interest that all traders together may hold in a stock's futures and options. Until September 2025 it was 20% of the shares held by non-promoters (the free float), and open interest was counted gross, with every option at its full notional value. From 1 October 2025 the limit is the lower of 15% of the free float and 65 times the average daily delivery volume, with a floor of 10% of the free float, recomputed every quarter, and open interest is counted in delta-adjusted (futures-equivalent) terms (Table 1). The new formula lowered every stock's limit, to a median of 73% of its old value.

The ratio of open interest to the limit is the stock's *utilisation*. When utilisation exceeds 95% at the end of a day, the stock is in the ban from the next day; the ban ends after utilisation falls to 80%. During the ban every participant may only reduce positions, in all expiries and in both futures and options, and rolling a position into the next month counts as a new position. Figure 2 shows the rule at work for SAIL, the stock banned most often.

How violations were detected changed twice, and this turns out to matter. Until 22 March 2020 an increase was judged on end-of-day positions, so a position opened and closed within the day was not a breach. From 23 March 2020, following SEBI's press release 18/2020, the clearing corporations judged violations on peak intraday positions. From 1 October 2025 each entity's end-of-day delta-adjusted position must fall relative to its position when the ban began, which allows new contracts that lower the delta, such as a put bought against a long future.

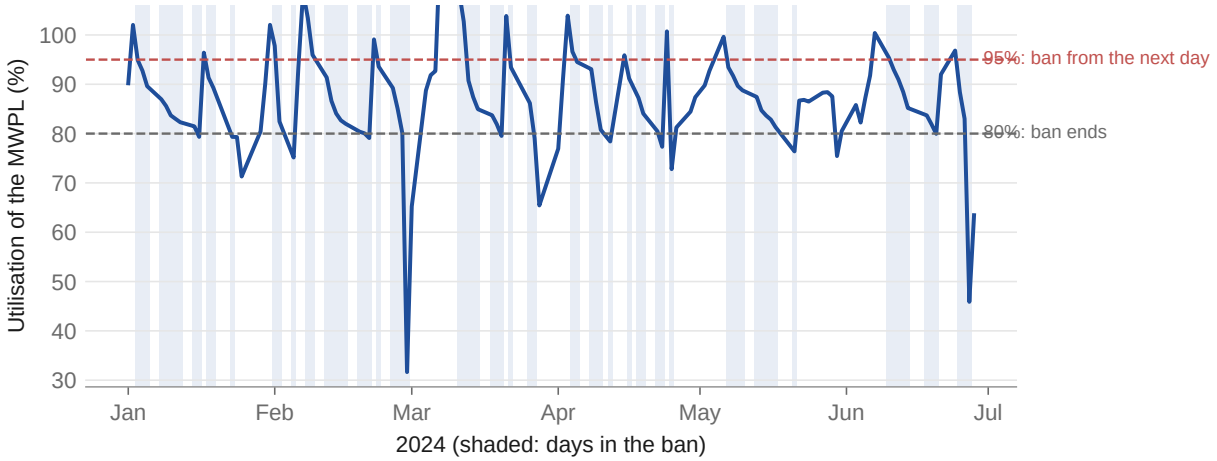
2.4 Trading costs and arbitrage capital

The STT on the sale of futures was 0.01% of the traded value from June 2013, 0.0125% from 1 April 2023, 0.02% from 1 October 2024 and 0.05% from 1 April 2026. A cash-and-carry position pays much more on the stock: 0.1% STT on the purchase and 0.1% on delivery at expiry, plus stamp duty, exchange charges and brokerage. The main holders of cash-and-carry positions

FIGURE 2

A stock enters the ban when open interest crosses 95% of its limit and leaves at 80%

Utilisation of the market-wide position limit for SAIL, January to June 2024; shaded days are ban days



Source: NSE combined open interest files and securities-in-ban lists; author's calculations.

are arbitrage mutual funds, which are taxed as equity funds. When debt funds lost their tax advantage on 1 April 2023, short-term money moved into arbitrage funds, and their assets grew from Rs 71,106 crore in April 2023 to Rs 2.98 lakh crore in September 2026.

3. RELATED LITERATURE

3.1 Why arbitrage is limited

The idea that arbitrage keeps prices in line rests on the arbitrageur being able to trade without limit and without risk. Shleifer and Vishny (1997) point out that arbitrage is done by specialists who invest other people's money. When a mispricing widens, the arbitrageur shows losses and investors withdraw funds, so capital leaves exactly when the opportunity is best. De Long, Shleifer, Summers and Waldmann (1990) show that the risk created by noise traders can by itself keep prices away from fundamentals, because an arbitrageur with a short horizon may have to close at a worse price. Gromb and Vayanos (2010) bring these mechanisms into one framework. Their account has three features that the model of this report shares: a demand shock moves the price by more when arbitrageurs are more risk averse or have less capital; a cost of short selling matters only when arbitrageurs need to be short, so the band around fair value is asymmetric; and margin requirements make capital a binding constraint. Short-sale constraints have a separate literature. Miller (1977) argues that when opinions differ and pessimists cannot sell short, prices reflect the optimists and are too high, and Diamond and Verrecchia (1987) show that such constraints slow the response of prices to bad news. These predictions are the subject of the second stage of the project.

3.2 The no-arbitrage band

A future and the stock bought with borrowed money give the same payoff at expiry, so the futures price should equal the cost of carry, and trading costs create a band around it within which arbitrage does not pay. Inside the band the basis should wander; outside, it should be pushed back. This calls for a threshold model. Dwyer, Locke and Yu (1996) estimate a three-regime threshold model for S&P 500 index futures with minute data and find thresholds of about half a round-trip cost and faster reversion outside the band than inside. Tse (2001) argues that arbitrageurs differ in their costs, so the market's response should strengthen smoothly with the size of the gap, and fits a smooth-transition model to Dow Jones futures. Ters and Urban (2020) show that in a threshold error-correction model the arbitrage cost is estimated precisely even when its components are not, and suggest comparing the band before and after regulatory changes. I use threshold and smooth-transition models from this tradition, applied to a daily panel of stocks rather than to a single index.

3.3 Evidence from India

Indian studies of the stock futures basis are few and use short intraday samples (Table 2). Aggarwal (2015) estimates threshold models for 141 stock futures with one-second order-book data over six months of 2013 and finds an average band of about −15 to +3 bp, wider for less liquid stocks, and moving with funding conditions. Shankar, Sankar and Kiran Kumar (2019) estimate weekly threshold models for 102 stock futures with five-minute data from 2007 to 2009 and find that both thresholds are negative, at about −39 and −10 bp, with the lower edge more sensitive to volatility, a sign of a constraint on the short side. Slivka, Shah and Wu (2011) add up the costs of arbitrage between a stock and its future and find a band about twice as wide as in mature markets. At the index level, Nandan, Agrawal and Bhargava (2014) find that Nifty futures traded below their cost-of-carry value on most days from 2003 to 2012. Saurav, Srivastava and Agarwalla (2026) study the stock futures basis around earnings announcements and short-sale constraints.

The intraday bands describe traders who open and close within minutes. The daily basis studied here measures the part of the gap that lasts overnight, where an arbitrageur must carry the position and pay the taxes on delivery, so it should be wider. To my knowledge no study has examined the F&O ban or modelled a position limit that, once reached, lets every participant only reduce positions. That is the gap this project fills, and this report lays the ground for it by measuring the band and the ban's effect on trading.

---

4. A MODEL OF ARBITRAGE UNDER INDIA'S LIMITS

Before looking at any data I wrote a model in which each Indian limit to arbitrage enters as a parameter. Its purpose is to say in advance what each limit should do, and what result would contradict it. This section presents the model and the propositions it implies; Appendix A gives the proofs.

TABLE 2

Earlier Indian estimates describe intraday bands; this report measures the overnight band

Studies of single-stock futures against their underlying stocks on the National Stock Exchange

STUDY	SAMPLE	DATA	BAND
Aggarwal (2015)	141 stocks, Mar to Aug 2013	One-second order book	About −15 to +3 bp, per stock and day
Shankar, Sankar and Kiran Kumar (2019)	102 stocks, 2007 to 2009	Five-minute prices	−39 to −10 bp, weekly
Slivka, Shah and Wu (2011)	Cost calculation	Fee schedules	About twice the width in mature markets
This report	360 stocks, 2015 to 2026	Daily volume-weighted prices	−34.5 to +23.9 bp around each stock's median

Source: the studies cited; Section 6 for this report.

4.1 Set-up

Let S be the price of a stock and F the price of its future, which expires at T. With an interest rate r and dividends whose present value is PV(D), the cost-of-carry value of the future is $F^* = (S - \mathrm{PV}(D))\, e^{r(T-t)}$, and the basis is

$$m = \ln F - \ln F^*, \tag{1}$$

measured in basis points of the stock price. A positive basis means the future is rich relative to the stock.

Two groups trade. *Outside traders*, such as speculators and hedgers, trade for their own reasons. Their net demand in the cash market is $D_S - \psi_S s$ and in the futures market $D_F - \psi_F f$, where s and f are the deviations of the stock and futures prices from fundamental value (so $m \approx f - s$), D_S and D_F shift demand, and ψ_S and ψ_F measure how deep each market is. An *arbitrage sector*, standing for arbitrage funds and proprietary desks, holds a units of the cash-and-carry trade: long the stock and short the future ($a < 0$ would be the reverse trade). Market clearing in both markets gives

$$m = \frac{u - a}{\Phi}, \qquad \frac{1}{\Phi} = \frac{1}{\psi_F} + \frac{1}{\psi_S}, \qquad u = \Phi\left(\frac{D_F}{\psi_F} - \frac{D_S}{\psi_S}\right). \tag{2}$$

The variable u is the *relative demand pressure* of outside traders toward the future: it is high when they want more exposure through futures than through the stock, for example when speculators seek leverage. Without arbitrage the basis would be u/Φ; every unit of cash-and-carry lowers it by $1/\Phi$.

The arbitrage sector chooses a to maximise

$$a\, m - \frac{\gamma}{2} a^2 - c_+ a^+ - c_- a^-, \qquad \gamma = \alpha \sigma^2, \tag{3}$$

TABLE 3

Each Indian limit to arbitrage enters the model as a parameter

Limits, their Indian form, and where they appear in the model

LIMIT	INDIAN FORM	PARAMETER
Trading costs	STT on futures and on delivery, stamp duty, fees, brokerage	Costs c_+, c_-
Short-sale constraints	Shorting only in F&O stocks; thin stock lending; the overnight short is a future	Cost of reverse trade c_-
Arbitrage capital	Arbitrage-fund assets; funding rates; margin rules	Risk aversion α, so γ
Arbitrage risk	Volatility of the basis before it closes	Risk σ^2, so γ
Position limits	MWPL; the F&O ban (enter above 95%, leave at 80%)	Reduce-only rule (Section 4.4)
Market depth	Liquidity of outside traders in each market	ψ_S, ψ_F, so Φ

Source: the model of this section.

where $a^{\pm} = \max(\pm a, 0)$. The first term is what the position earns when the basis closes at expiry. The second is the cost of risk: α is risk aversion per unit of capital, which is higher when arbitrage capital is scarce, and σ^2 is the risk that the basis widens before it closes. The last two terms are the costs of each trade: c_+ for cash-and-carry (taxes, fees, price impact, and any gap between the return arbitrage capital requires and the rate used to compute the basis) and c_- for the reverse trade, which also pays for borrowing the stock and which is very high in India because overnight short sales are hard. Table 3 lists how each Indian limit enters.

4.2 The no-arbitrage band

Proposition 1 (An asymmetric band)**.** *The equilibrium basis is unique and equals*

$$m(u) = \begin{cases} \dfrac{\gamma u + c_+}{\gamma\Phi + 1}, & u > \Phi c_+, \\ \dfrac{u}{\Phi}, & -\Phi c_- \le u \le \Phi c_+, \\ \dfrac{\gamma u - c_-}{\gamma\Phi + 1}, & u < -\Phi c_-. \end{cases} \tag{4}$$

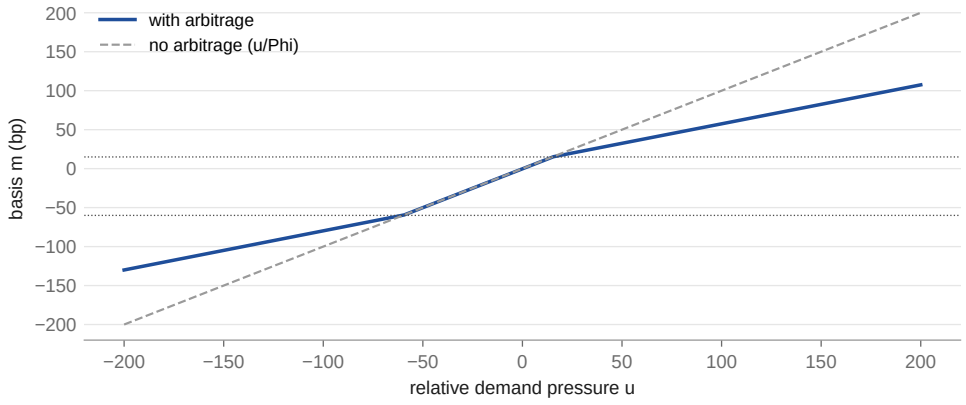
In words: as long as the basis lies between $-c_-$ and c_+, no arbitrage pays, and the basis simply follows the demand of outside traders. Once it moves beyond an edge, arbitrage starts and absorbs part of each further push, so the basis moves less (Figure 3). Because the reverse trade is costlier, the band reaches further below zero than above it.

Proposition 2 (What moves the band)**.** *(i) The edges of the band are c_+ and $-c_-$: costs set the edges, and a change in a tax moves the corresponding edge one for one. (ii) Capital and*

FIGURE 3

Inside the band the basis follows demand one for one; outside it arbitrage absorbs part of each push

Equilibrium basis against the demand pressure of outside traders, with illustrative costs $c_+ = 15$ bp and $c_- = 60$ bp



Source: Proposition 1; illustrative parameters, not estimates.

> *risk do not move the edges; they set how far the basis travels beyond them, since the slope $\gamma/(\gamma\Phi + 1)$ rises with γ. (iii) The carrying part of the costs is proportional to the time left to expiry, so the band narrows as expiry approaches. (iv) The band is centred at $(c_+ - c_-)/2$, below zero whenever the reverse trade costs more than cash-and-carry.*

Part (i) gives a direct test. When the STT on futures was raised, the cost of cash-and-carry rose, so the upper edge of the band should have risen, and by more for larger increases.

4.3 Existing positions

Arbitrageurs do not start each day from zero. Suppose the sector enters the day holding a_{t-1} units and pays c_u per unit to unwind them.

> **Proposition 3** (The band moves with the existing book)**.** *The sector adds to its position only if $m_t > \gamma a_{t-1} + c_+$, unwinds only if $m_t < \gamma a_{t-1} - c_u$, and otherwise holds its position. The band within which it does not trade is* $[\gamma a_{t-1} - c_u,\ \gamma a_{t-1} + c_+]$.

The edges of the band are therefore not fixed: they shift with the size of the existing book. A threshold model with fixed thresholds, the standard tool, will estimate an average position of the edges rather than the costs themselves. I checked this before using real data: when I simulated the basis from this model with costs of 15 bp to add and 10 bp to unwind, the threshold model placed the thresholds at about -10 and 0 bp, so the estimated upper threshold lay well below the 15 bp cost of adding to a position. This is why my test of costs compares how the edges *change* around tax increases rather than comparing their level with the costs.

4.4 The F&O ban

The ban acts on everyone in the market, so the model represents it with two restrictions.

Assumption 1 (The ban)**.** During a ban, (B1) the arbitrage sector may only reduce its position, $0 \leq a_t \leq a_{t-1}$; and (B2) outside traders cannot open new futures positions. A share χ of this blocked demand moves to the stock (a trader who cannot buy the future buys the share instead) and the rest is withdrawn. Outside traders can still close existing positions.

> **Proposition 4** (One-sided correction)**.** *During a ban, rich futures are no longer corrected: since nobody may open new cash-and-carry positions, any further pressure toward the future passes through to the basis in full. Cheap futures are still corrected, by arbitrageurs unwinding their positions, as long as they have positions left to unwind. This correction weakens as positions run down, and an expiry inside the ban removes the expiring positions, since they cannot be rolled.*

> **Proposition 5** (The level effect has no fixed sign)**.** *Compared with normal times, a ban raises the basis when the pressure on a stock comes mainly from traders closing positions, and lowers it when new demand keeps arriving, because part of the blocked demand for futures moves to the stock. Which case applies depends on the composition of trading before the ban.*

> **Proposition 6** (Trading in a ban)**.** *If the rule binds on every trade, each futures trade during a ban closes a long position and a short position. The change in open interest then equals minus the volume traded, so the ratio $R = (\Delta OI + V)/V$, which is close to one on normal days, is zero, and futures volume falls. Trades that offset positions across instruments, which are allowed from October 2025, break this identity.*

Figure 4 shows a simulated ban. When the ban starts, the arbitrage book can only shrink, and the basis follows demand upward without resistance. One further implication concerns the days before a ban. If closing a position is costlier during a ban, an arbitrageur who opens a position when utilisation is already high must allow for the chance of being caught in one, so the upper edge of the band should rise as utilisation approaches 95%. The model does not, however, say whether utilisation should bunch just below the trigger: arbitrageurs holding back would make it thinner there, while speculators rushing in before a ban would do the opposite.

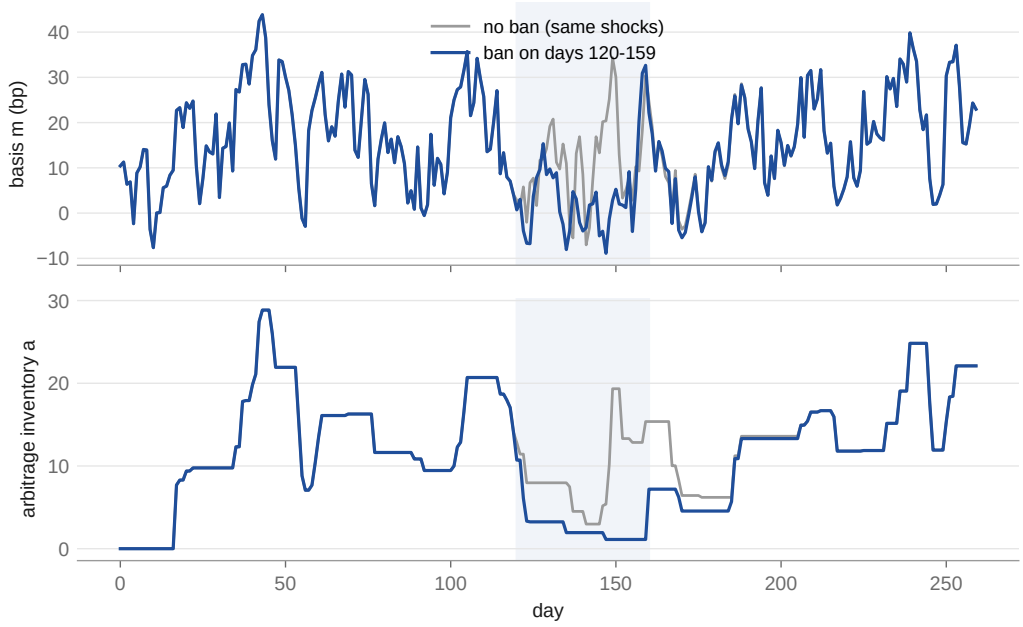
4.5 From the model to the tests

Table 4 lists the predictions, the result that would contradict each, and where it is tested. The predictions about the band, about costs and frictions, and about trading in a ban are tested in this report. The predictions about what the ban does to the level and the correction of the basis, and the model's predictions about capital and short-sale constraints, are tested in the second stage. The labels in the last column are those of the pre-analysis plan, which names four primary hypotheses (H3, H6, H7, H8) and corrects them jointly for multiple testing.

FIGURE 4

In a ban the arbitrage book can only shrink, so upward pressure on the basis passes through in full

One simulated path with and without a ban on days 120 to 159, same shocks; basis (top) and arbitrage position (bottom)



Source: the model of this section, simulated; illustrative parameters, not estimates.

5. RESEARCH DESIGN AND DATA

5.1 How the work was organised

A common risk in empirical work is that the hypotheses, the specification and the data are adjusted to each other after the results are seen. To limit this, the work followed a fixed sequence (Figure 5). The model came first. Its predictions, the samples, the variable definitions, the estimating equations and the results that would count against each prediction were then written into a pre-analysis plan, which I froze with a time-stamped tag in the project's version control on 5 November 2026. Until then I had looked only at the data on bans and position limits and at the basis on its own, never at how one related to the other. Where the plan left a detail open, I wrote the choice into a dated log before running the test that needed it. Each estimator was first run on data simulated with a known effect, to check that it recovers that effect. Every pre-specified test is reported, including those that go against the model.

The plan names four primary hypotheses: that costs set the edges of the band (H3), that trading leaves futures in a ban (H8), that correction becomes one-sided in a ban (H7), and that access to short selling speeds price adjustment (H6). Their p-values are corrected jointly by Holm's method at the 5% level. The other tests are secondary and are reported without adjustment. This report presents H3 and H8 together with the secondary tests that belong to the same questions;

TABLE 4

Every prediction comes with the result that would contradict it

Predictions of the model, and where each is tested; primary hypotheses of the test plan in bold

PREDICTION	CONTRADICTED IF	TESTED IN	PLAN LABEL
THIS REPORT			
The basis moves in an asymmetric band, wider below zero (Props. 1, 2)	No band; a symmetric band	Section 6	Band estimation
Tax increases raise the upper edge, more for larger increases (Prop. 2)	Edges unrelated to cost changes	Section 7	**H3**
Bands are wider where frictions are higher (Props. 1 to 3)	No relation	Section 7	H1
In a ban futures volume falls and $\Delta OI \approx -V$ (Prop. 6)	Futures activity unchanged	Section 8	**H8**
Utilisation near the 95% trigger (not signed)	(descriptive)	Section 8	Density test
SECOND STAGE			
In a ban rich futures revert more slowly (Prop. 4)	No asymmetry, or the opposite	Stage II	**H7**
The level effect follows the composition of trading (Prop. 5)	Sign unrelated to composition	Stage II	Level effect
The upper tail rises as utilisation nears 95%	A jump only at 95%	Stage II	H9
Scarcer capital lets the basis travel further (Prop. 2)	No relation	Stage II	H4
Access to short selling speeds price adjustment and lowers overpricing	No change, or the opposite	Stage II	**H6**, H5, H2

Source: the model of this section; the pre-analysis plan (Appendix B).

H6 and H7 belong to the second stage.

5.2 Data

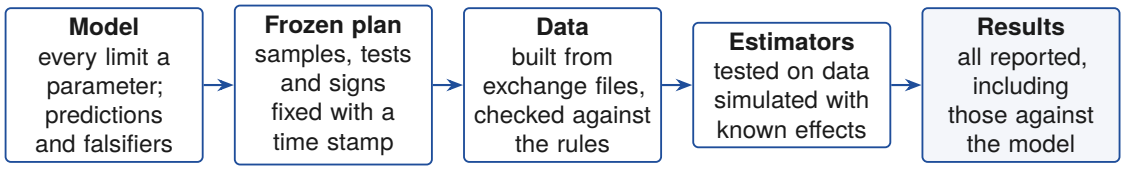
All data come from public archives of the NSE and are downloaded and processed by scripts, so every step can be repeated (Table 5). The main files are the daily combined open-interest files, which give each stock's MWPL and open interest from February 2010; the daily lists of stocks in the ban; and the daily price files (bhavcopies) of the futures and cash markets from 2015 to October 2026.

Measuring the basis. For each stock and day I compute the basis (1) from the volume-weighted average prices of the future and of the stock, so that both prices come from trades over the same

FIGURE 5

Theory and tests were fixed before any outcome was related to any treatment

The sequence of the work, from model to results



The plan and its log are described in Appendix B.

TABLE 5

Every series comes from a public archive

Sources, content and coverage of the data used in this report

SOURCE	CONTENT	COVERAGE
NSE combined open interest files	MWPL, open interest, futures-equivalent open interest (from Oct 2025)	Feb 2010 to Oct 2026
NSE securities-in-ban lists	Official daily list of stocks in the ban	Feb 2010 to Oct 2026
NSE F&O and cash bhavcopies, delivery files	Prices, traded value and quantity, contracts, open interest, delivery	2015 to Oct 2026
NSE corporate actions	Dividends, splits, bonuses and other events with ex-dates	2014 to 2026
NSE SLB bhavcopies	Daily stock-lending trades	2015 to Oct 2026
OECD via FRED	Three-month interbank rate (monthly)	2011 to 2026

Source: National Stock Exchange of India archives; OECD Main Economic Indicators via FRED.

day:

$$m_{it} = \ln \bar{F}_{it} - \ln\left(\bar{S}_{it} - \mathrm{PV}_{it}(D)\right) - r_t \tau_{it}, \tag{5}$$

where τ_{it} is the time to expiry in years and r_t the three-month interbank rate. I use the nearest contract and switch to the next one when three or fewer trading days remain, require at least five contracts traded, and drop contracts with a dividend or other corporate event during their life (where the adjustment is uncertain) and the first 20 trading days after a stock enters the F&O segment. To limit the influence of data errors, the basis is winsorised at the 1st and 99th percentiles within each year. The final sample has 486,040 stock-days for 371 stocks, 8,851 of them in a ban.[1]

[1]The data were checked against the exchange's own rules before use. The "no fresh positions" flags in the open-interest files match the next day's official ban list for 11,753 of 11,764 flags; utilisation was above 95% on the day before 99.4% of ban spells began and at or below 80% on the last day of 99.7%; and the futures volume-weighted price lies within the day's high and low for 99.8% of contracts. Appendix B.2

TABLE 6

Bans are frequent, concentrated in a few stocks, and longer under the new rule

F&O stocks, banned stocks, ban spells and ban stock-days by year

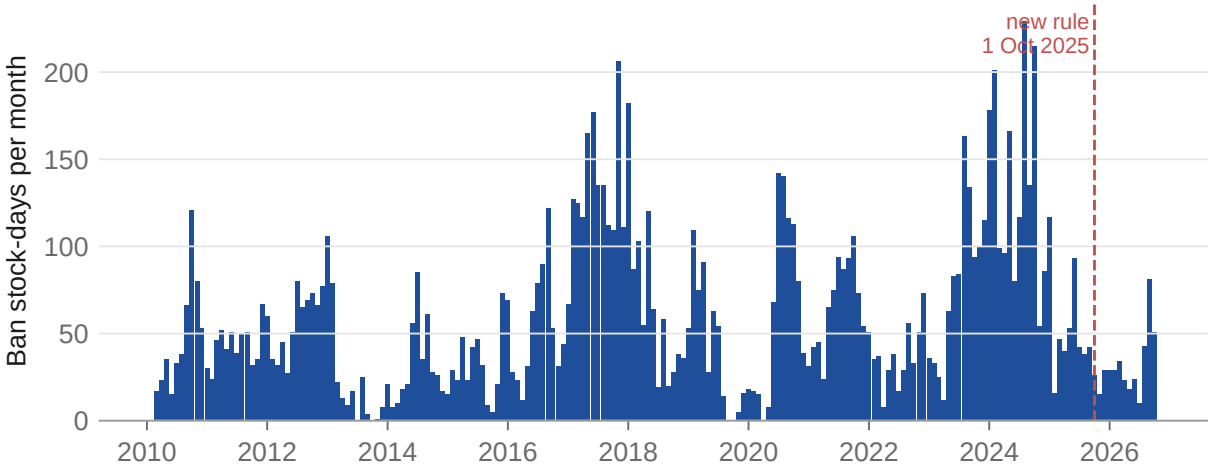
Year	F&O stocks (avg per day)	Stocks banned	Ban spells	Ban stock-days	Median spell (days)	Longest spell (days)
2015	154	16	69	367	4	21
2016	172	27	136	645	4	17
2017	200	43	287	1586	5	24
2018	206	38	162	810	4	31
2019	176	23	86	508	4	17
2020	140	40	178	756	3	13
2021	163	26	162	789	4	16
2022	196	23	100	457	4	13
2023	187	41	193	942	4	17
2024	185	49	289	1656	5	22
2025	216	37	93	558	5	16
2026[a]	209	10	33	342	10	29
2015 to 2026	183	141	1788	9416	4	31

[a] January to 27 October 2026. Spells are counted in the year they start. Source: NSE securities-in-ban lists and F&O bhavcopies; author's calculations.

FIGURE 6

Bans come in waves and became rarer under the new rule

Number of stock-days in the ban per month, February 2010 to October 2026



Source: NSE securities-in-ban lists; author's calculations.

5.3 What the data show

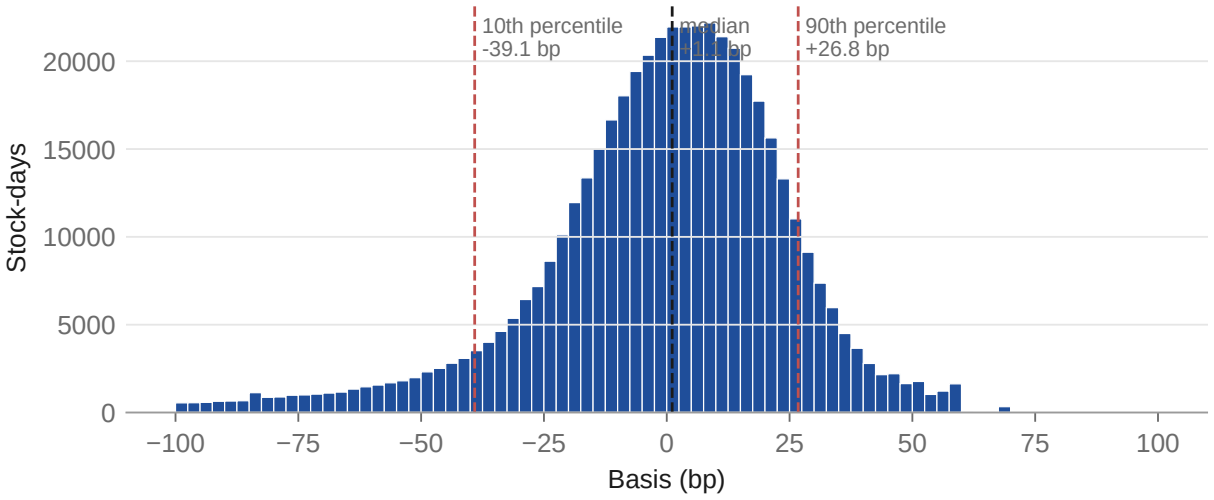
Bans. From 2015 to October 2026 there were 1,788 ban spells, covering 9,416 stock-days and 141 stocks (Table 6). Bans are concentrated in a few stocks and come in waves (Figure 6). A typical spell lasted four trading days before the October 2025 reform; the 44 spells that began after it have a median of nine.

gives more detail.

FIGURE 7

The basis is centred near zero with a long lower tail

Basis of the analysis sample, 2015 to 2026, in basis points (3.1% of stock-days lie beyond 100 bp from zero and are not drawn)

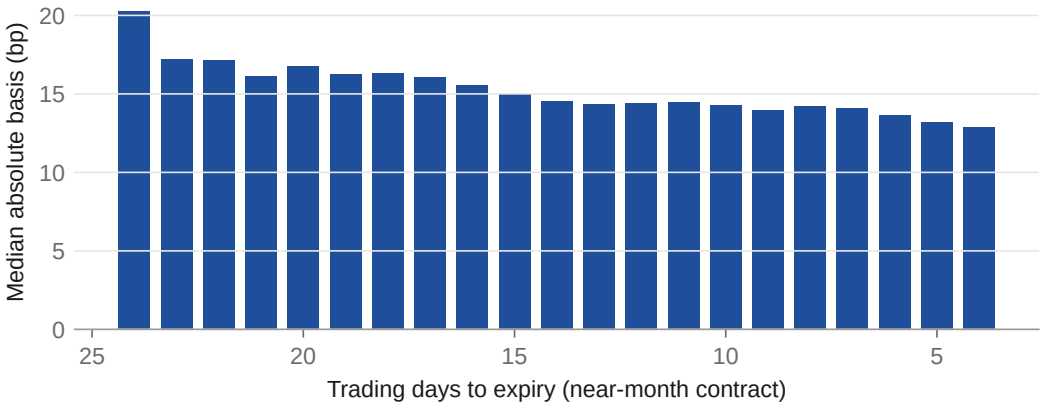


Source: NSE F&O and cash bhavcopies, corporate actions, three-month interbank rate; author's calculations.

FIGURE 8

The basis narrows as expiry approaches

Median absolute basis of the near-month contract by trading days to expiry (in the last three days the sample uses the next contract)



Source: as for Figure 7.

The basis. The median basis is +1.1 bp, so on a typical day futures are priced very close to their cost-of-carry value. The distribution is skewed: the 10th percentile is −39.1 bp and the 90th +26.8 bp (Figure 7). Cheap futures are thus more common and further from zero than rich ones, which is what one expects when the trade that would correct cheap futures, short selling the stock, is the costlier one. The basis also narrows as expiry approaches (Figure 8), as the carrying part of the costs shrinks with the time left.

6. THE NO-ARBITRAGE BAND

Proposition 1 says that the basis should behave differently inside and outside a band: inside, it follows the demand of outside traders and nothing pulls it back; outside, arbitrage pushes it toward the band. This section estimates where the band lies, using days on which the stock is not in a ban.

6.1 A threshold model

Let x_{t-1} be yesterday's basis minus the stock's median basis on non-ban days, in basis points, and Δm_t the change in the basis of the same contract from yesterday to today. Measuring the basis relative to each stock's own median removes permanent differences between stocks, such as a stock that always trades slightly rich. The three-regime threshold model is

$$\Delta m_t = a_r + \rho_r\, x_{t-1} + e_t, \qquad r = \begin{cases} \text{lower} & x_{t-1} < g_L, \\ \text{middle} & g_L \le x_{t-1} \le g_U, \\ \text{upper} & x_{t-1} > g_U. \end{cases} \tag{6}$$

The thresholds g_L and g_U are the estimated edges of the band, and ρ_r measures how strongly the basis moves back in each regime: $\rho_r = 0$ means no pull at all and $\rho_r = -1$ means that a deviation disappears within a day. The model predicts ρ close to zero in the middle regime and clearly negative outside it. I choose the two thresholds by least squares over a grid of percentiles of x, keeping at least 10% of the observations in each outer regime and 20% in the middle. Because the thresholds are chosen from the data, the usual F-test against a linear model is not valid; I compare the largest F-statistic over the grid with its distribution under the linear model, obtained by a wild bootstrap clustered by trading day. Confidence intervals for the thresholds come from resampling whole stocks.

6.2 Estimates

The sample has 473,110 stock-days for 360 stocks. Linearity is rejected decisively: the test statistic is 7,471 against a 95% critical value of 31. The band runs from -34.5 bp to +23.9 bp around each stock's median, about 58 bp wide, and both edges are estimated precisely (Table 7). The basis lies inside the band on 76% of days and in each outer regime on 12%. As the model predicts, the band is asymmetric: the lower edge lies further from the centre than the upper edge. Figure 9 shows the pattern directly: the further the basis is from the stock's median yesterday, the more it moves back today, with the strongest pull above the upper edge.

6.3 How much of the reversion is noise

The slopes need care. Inside the band the model predicts almost no pull, yet the middle slope of -0.47 implies that half of a deviation disappears within about a day. The likely reason is measurement noise. The daily volume-weighted prices of the future and the stock are averages

TABLE 7

The basis moves in a wide, asymmetric band, but much of the daily reversion is noise

Pooled three-regime threshold model, non-ban days, 2015 to 2026 (473,110 observations, 360 stocks)

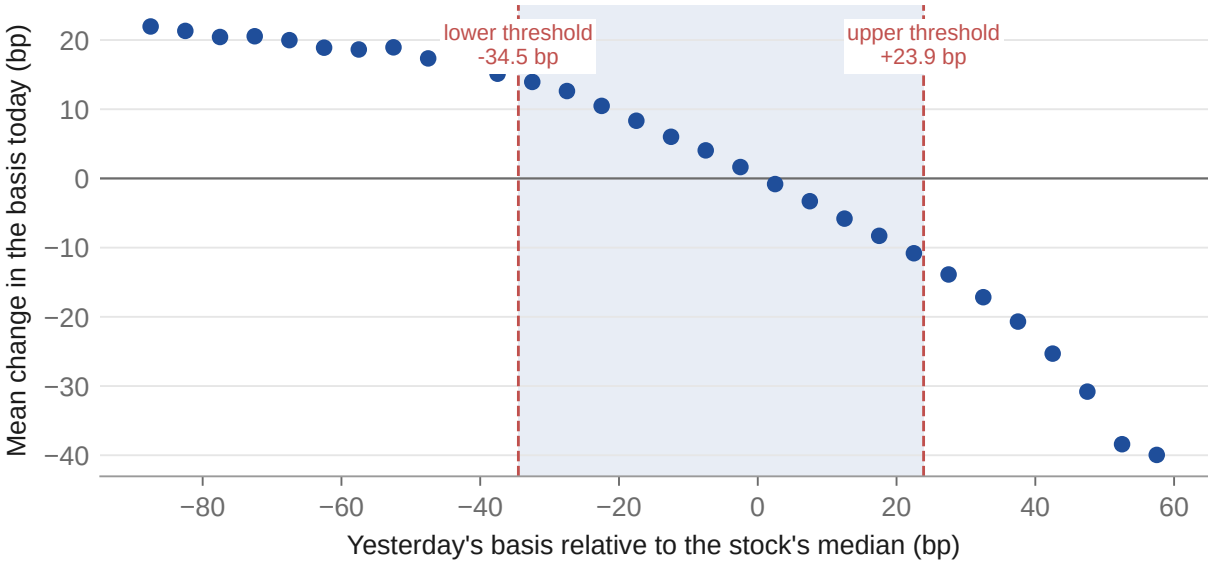
	LOWER REGIME	MIDDLE REGIME	UPPER REGIME
Threshold (bp, relative to the stock's median)	below −34.5		above +23.9
95% interval (stock bootstrap)	[−37.2, −32.2]		[+20.3, +26.1]
Share of observations	0.12	0.76	0.12
Slope ρ_r	−0.084	−0.468	−0.809
Slope, yesterday's deviation instrumented	−0.121	−0.192	−0.141

Thresholds chosen by least squares over a grid of percentiles; intervals from resampling whole stocks. In the last row yesterday's deviation is instrumented by the deviation two days earlier, and regimes are defined on that deviation. Source: author's calculations.

FIGURE 9

The further the basis moves from the stock's median, the more it moves back the next day

Mean change in the basis today by yesterday's deviation from the stock's median (5 bp bins), non-ban days; shaded: the estimated band



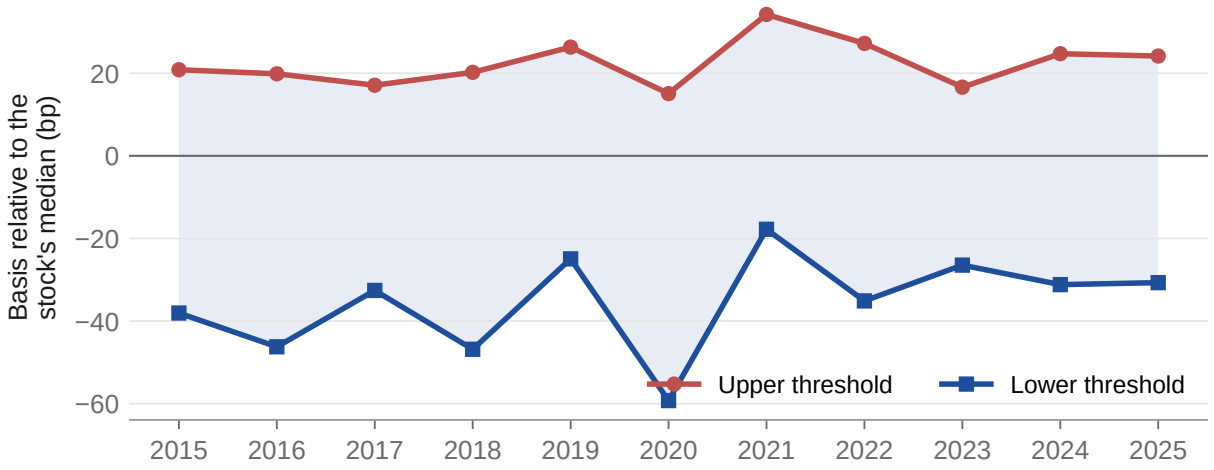
Source: author's calculations.

over slightly different trades, so the measured basis contains an error that does not persist. An error in yesterday's basis appears in x_{t-1} with one sign and in Δm_t with the other, which pushes the estimated slope toward −1 even if the true basis does not move back at all. A standard remedy is to instrument yesterday's deviation with the deviation two days earlier, whose error is unrelated to yesterday's. With this correction all three slopes become small and similar (−0.12, −0.19 and −0.14, last row of Table 7). Much of the day-to-day reversion is therefore noise,

FIGURE 10

The band is stable from year to year, apart from the dislocation of March 2020

Thresholds of the pooled threshold model by calendar year, 2015 to 2025, non-ban days, basis relative to each stock's median



Years with a change in the tax are examined in Section 7. Source: author's calculations.

and the faster pull above the upper edge in the uncorrected model is not a robust finding. The thresholds, which describe where the basis spends its time rather than how fast it returns, are the more reliable part of the estimates.

6.4 Smooth adjustment

Arbitrageurs differ in their costs, so the market's response may strengthen gradually rather than switch on at one threshold (Tse, 2001). I therefore also estimate smooth-transition models of the form

$$\Delta m_t = a + \left[\rho_1 + \rho_2\, G(x_{t-1})\right] x_{t-1} + e_t, \tag{7}$$

where G moves smoothly from 0 to 1. With an exponential G, which is symmetric around a centre, deviations in either direction are treated alike; with a logistic G the pull can differ on the two sides. By the Bayesian information criterion the threshold model fits best, the logistic model almost as well (with weak reversion below about +5 bp and strong reversion above), and the symmetric exponential model clearly worse. A symmetric model does not suit an asymmetric band.

6.5 The band over time and over the life of a contract

Estimated year by year, the band is stable (Figure 10). The exception is 2020, when the market dislocation of March pushed the lower edge to −59 bp. The band also depends on the time left to expiry, as Proposition 2 (iii) predicts: the lower edge is at −40 bp when 17 or more trading days remain and at −28 bp in the last ten days. In a smooth-transition model in which the strength of reversion depends on the days left to expiry, reversion is stronger in the last week of a contract.

6.6 Bands of individual stocks

Estimating the model separately for each of the 258 stocks with at least 500 non-ban days gives a median band of −26 to +17 bp, about 41 bp wide, with large differences between stocks: the middle 80% of stocks have widths between about 18 and 76 bp. These bands are much wider than the intraday bands of earlier Indian studies (Table 2), which is expected: an arbitrageur who holds a position overnight pays the taxes on delivery and bears the risk of carrying it, while an intraday trader does neither. Section 7 asks which costs and frictions explain the location of the band and its differences across stocks.

---

7. WHAT SETS THE BAND

Proposition 2 says that trading costs set the edges of the band. This section tests that prediction in two ways: over time, using the three increases in the tax on futures, and in levels, by adding up what an arbitrage position actually costs. It then asks whether the band is wider for stocks that are harder to trade.

7.1 Tax increases (pre-specified test H3)

The STT on the sale of futures rose on 3 April 2023 (the first trading day after the 1 April change), 1 October 2024 and 1 April 2026, by 0.25, 0.75 and 3.0 bp of the futures sale. Each increase raises the cost of cash-and-carry, so the upper edge of the band should rise, and by more for larger increases. Because Proposition 3 implies that fixed thresholds measure an average edge rather than the cost, the plan tests the *change* in the upper part of each stock's basis distribution. For stock i and event e, let $q^{80}_{ie,\text{pre}}$ and $q^{80}_{ie,\text{post}}$ be the 80th percentiles of its basis in the 40 trading days before and the 40 from the event. The test statistic is the average change

$$\overline{\Delta q}^{80} = \frac{1}{N} \sum_{i,e} \left(q^{80}_{ie,\text{post}} - q^{80}_{ie,\text{pre}} \right), \tag{8}$$

over the 2024 and 2026 events, with standard errors clustered by stock. The 2023 event is reported but not used as evidence, because the taxation of debt funds changed on the same day, which may have moved arbitrage capital at the same time.

The upper percentile rose after both later increases, by 6.4 bp in 2024 and 7.4 bp in 2026, and fell slightly in 2023 (Table 8). The pooled change is +6.96 bp (standard error 0.61), and the three events are ordered as the cost changes are. On the test fixed in advance, H3 is supported, and it remains significant after the Holm correction for the four primary hypotheses.

Two features of the result call for caution. First, the rises are larger than the cost changes: an increase of 0.75 bp in tax went with a 6.4 bp rise in the upper percentile. Second, the median and the 20th percentile also rose in 2024 and 2026, so the whole distribution shifted rather than just its upper edge. Such shifts can come from anything that affects all stocks at once, and a standard error clustered only by stock does not allow for them, because every stock shares the

TABLE 8

The upper part of the basis rose after the 2024 and 2026 tax increases

Mean change in each stock's percentiles of the basis, 40 trading days after the event against 40 before (bp)

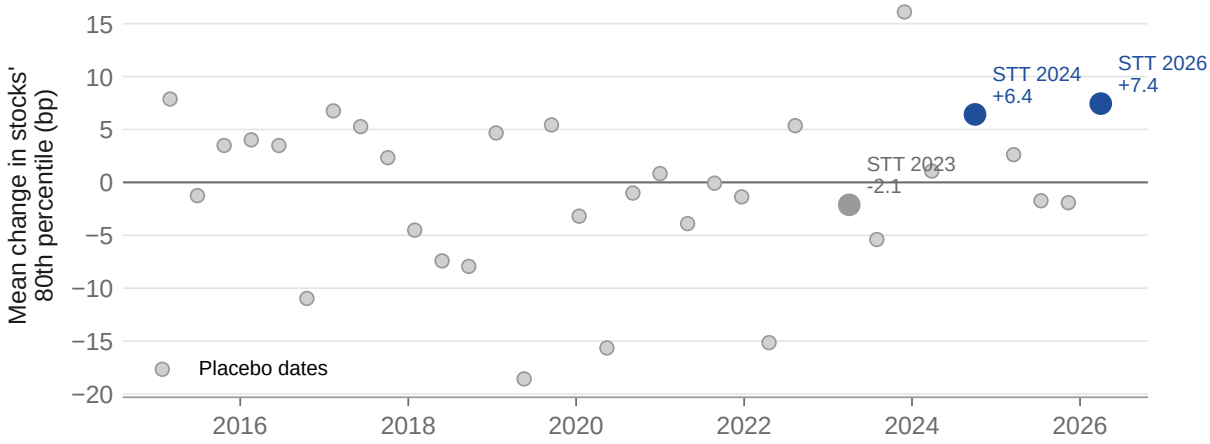
EVENT	STT CHANGE	STOCKS	80TH PERCENTILE	20TH	MEDIAN
3 Apr 2023[a]	+0.25	181	−2.12 (0.87)	+0.24	+0.27
1 Oct 2024	+0.75	172	+6.42 (0.70)	+4.55	+5.39
1 Apr 2026	+3.0	192	+7.44 (0.98)	+2.10	+6.30

Standard errors in brackets. STT change in bp of the futures sale. [a] Same day as the change in the taxation of debt funds; reported, not used as evidence. Source: author's calculations.

FIGURE 11

Against placebo dates, the rise after the tax increases is large but not exceptional

Mean change in stocks' 80th percentile of the basis at the three STT dates and at 30 placebo dates (bp)



Placebo dates every 80 trading days, away from the true events. Source: author's calculations.

same event date (Kolari and Pynnönen, 2010). The plan therefore also specified a placebo check: the same statistic at 30 dates away from the true events. Across these dates it has a mean of −1.0 bp and a standard deviation of 7.4 bp, and 10% of placebo dates show a rise at least as large as in 2024, 7% one as large as in 2026 (Figure 11). Judged against what happens on ordinary dates, the rises after the tax increases are large but not exceptional. My reading is that costs push the band in the predicted direction but do not set its edges as tightly as the model implies.[2]

7.2 What an arbitrage position costs

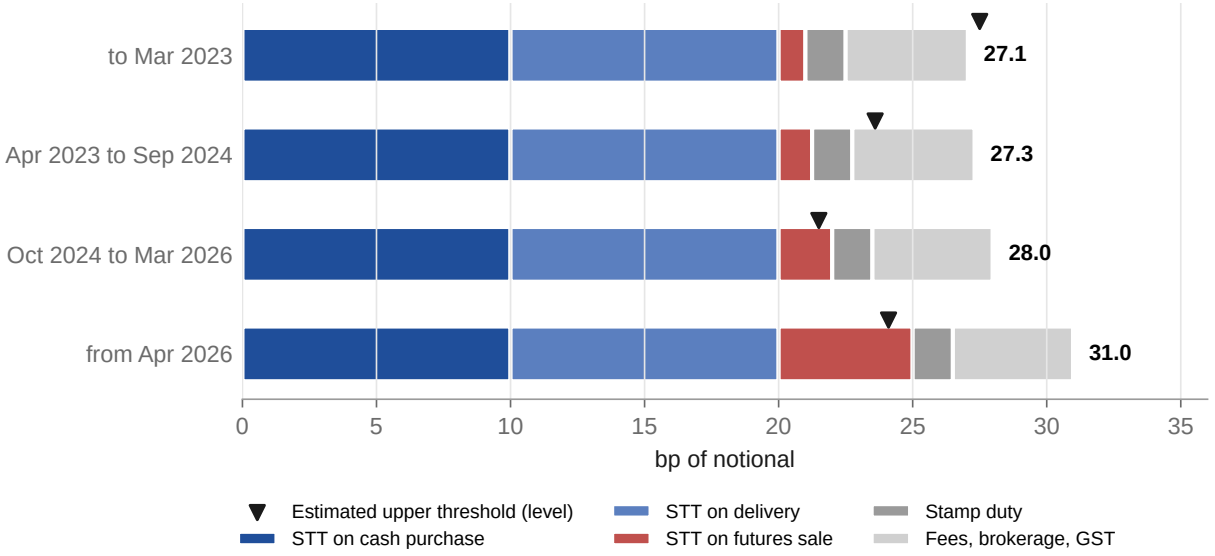
A second way to compare the band with costs is to add up the cost of an actual position. An overnight cash-and-carry position held to delivery pays STT of 0.1% on the purchase of the stock and again on delivery at expiry, the STT on the futures sale, stamp duty, exchange charges, the SEBI fee, goods and services tax on fees, and brokerage. With an institutional brokerage

[2]The same statistic at the three dates of SEBI's 2024 to 2025 reforms of index derivatives, which did not change the cost of single-stock arbitrage, gives −4.1, −0.3 and +1.9 bp, all within the placebo range.

FIGURE 12

The full cost of a new position lies above the estimated upper edge

Round trip of an overnight cash-and-carry position held to delivery, by STT period (bp of notional); triangles: estimated upper threshold in levels



Institutional brokerage of 0.01% per leg; GST of 18% on fees and brokerage. Source: NSE circulars on STT and transaction charges; author's calculations.

of 0.01% per leg, the full round trip costs 27.1, 27.3, 28.0 and 31.0 bp in the four tax periods (Figure 12), or 25 to 38 bp for brokerage between 0.005% and 0.03% per leg. Rolling an existing position for one more month is far cheaper, 4 to 8 bp, because the delivery taxes are paid only once.

The estimated upper threshold of the basis, measured in levels rather than relative to each stock's median, lies between +22 and +28 bp in the four periods, below the full cost of a new position, which falls between the 90th and 95th percentiles of the basis. This is what Proposition 3 implies when arbitrageurs already hold positions. New capital enters only when the gap is large enough to pay the full round trip, while funds that keep their capital in place and roll it from month to month can lean against much smaller gaps. The edge measured in the data is a mixture of the two.

7.3 Frictions across stocks (H1)

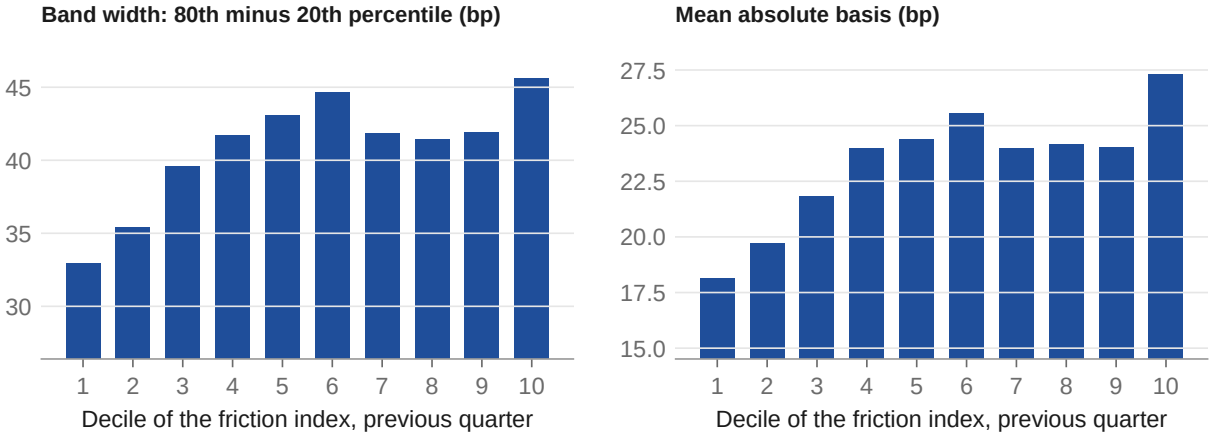
The band should also be wider for stocks in which arbitrage is harder. For each F&O stock and quarter I build a friction index from five measures taken in the previous quarter: the bid-ask spread estimated from daily high and low prices (Corwin and Schultz, 2012), illiquidity (Amihud, 2002), idiosyncratic volatility, utilisation of the position limit, and the share of days with no stock lending. The index is the first principal component of these measures, standardised within each quarter. I then regress the width of each stock's band in the quarter (the 80th minus the 20th percentile of its basis on non-ban days) on the index, with quarter fixed effects and standard errors clustered by stock and quarter.

A stock with an index one standard deviation higher has a band 3.5 bp wider (standard error

FIGURE 13

Bands are wider in stocks that are harder to trade

Band width and mean absolute basis by decile of the friction index of the previous quarter, F&O stocks, 2015 to 2025 (quarter means removed)



Index: first principal component of spread, illiquidity, idiosyncratic volatility, utilisation of the position limit and lack of stock lending. 7,398 stock-quarters, 347 stocks. Source: author's calculations.

1.1, one-sided $p = 0.002$; the mean width is 40.8 bp) and a mean absolute basis 2.5 bp larger (Figure 13). Most of the difference is between the least frictional stocks and the rest. The relation holds across stocks, not within a stock over time: with stock fixed effects it disappears, so it reflects persistent differences between stocks rather than changes in a stock's frictions. The index also does not predict how fast the basis returns to the band, which is consistent with the finding above that daily reversion is mostly noise. One component enters with the opposite sign: stocks with fewer lending days have narrower bands, which suggests that lending activity measures demand to borrow more than the ease of borrowing.

8. DOES THE BAN BIND ON TRADING?

Before asking what a ban does to prices, which is the subject of the second stage, I first ask whether the rule changes trading at all. A rule that lets traders only reduce positions should have three visible effects (Proposition 6): futures trading should fall, some of the blocked demand should move to the cash market, and every futures trade should close existing positions. This section tests these effects, looks at what happens at the 95% trigger itself, and measures how the October 2025 rules changed the number of bans.

8.1 Trading on ban days (pre-specified test H8)

A simple comparison of ban days with other days would mix the effect of the ban with the effect of the heavy trading that pushed the stock into it. The plan therefore compares each stock's ban days with its own non-ban days at a similar level of utilisation the day before. Between 80% and 95% both kinds of days occur: a stock that crossed 95% stays in the ban until it falls to 80%,

TABLE 9

In a ban, futures trading collapses and cash trading moves toward delivery

Ban days against the same stock's non-ban days at utilisation of 80% to 95% the day before, January 2015 to September 2025

OUTCOME	EFFECT OF A BAN (S.E.)	MEAN ON NON-BAN DAYS
log(1 + futures contracts traded)	−2.00 (0.11)	8.50
log(1 + cash shares traded)	+0.073 (0.027)	15.80
Delivery share of cash volume	+0.041 (0.005)	0.26

Equation (9): stock-by-utilisation-bin and date fixed effects; standard errors clustered by stock and date; expiry days excluded. 16,270 stock-days, 5,441 of them in a ban, 167 stocks. Source: author's calculations.

while a stock that has not crossed is free. The regression is

$$y_{it} = \beta \, \text{Ban}_{it} + \alpha_{i,b(\omega_{i,t-1})} + \lambda_t + \varepsilon_{it}, \tag{9}$$

where Ban_{it} marks ban days, $\alpha_{i,b}$ is a fixed effect for each stock and each 0.05-wide bin of yesterday's utilisation $\omega_{i,t-1}$, and λ_t is a fixed effect for each day, which absorbs market-wide news. Standard errors are clustered by stock and by day. The sample covers stock-days before October 2025 with utilisation between 80% and 95% the day before, excluding expiry days, when open interest falls mechanically.

In a ban, futures trading falls by 2.0 log points, or about 87%. Cash-market volume rises by about 7.5%, and the share of cash trading that ends in delivery rises by 4.1 percentage points from an average of 26% (Table 9). All three effects have the predicted sign. H8 is supported, also after the Holm correction for the four primary hypotheses.

8.2 Does every trade close a position?

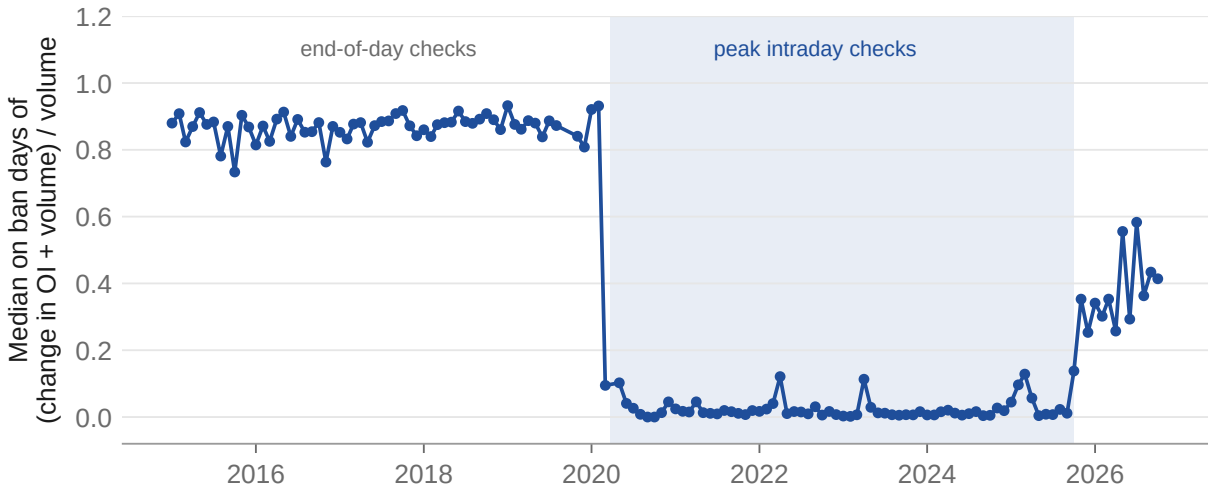
If the rule binds on every trade, each futures trade during a ban must close a long position on one side and a short position on the other, so open interest falls by exactly the volume traded. The ratio $R = (\Delta OI + V)/V$ is then zero, while on a normal day, when most trades open new positions, it is close to one. The plan set the reading in advance: the identity holds if the mean ratio on ban days is close to zero. Over the whole period it is not: the mean on ban days is 0.42.

The reason became clear once the ratio is plotted over time (Figure 14). It changed abruptly in March 2020, at exactly the point when the enforcement of the rule changed. Until 22 March 2020 violations were judged on end-of-day positions, so a trader could open and close a position within the day; the median ratio on ban days was then 0.87, close to a normal day. From 23 March 2020 violations were judged on peak intraday positions, and the median fell to 0.011: almost every trade closed a position, as the model predicts when the rule binds trade by trade. Under the rule of October 2025, which allows any trade that lowers an entity's end-of-day delta, the median rose again to between 0.19 and 0.36. The fall in futures trading itself was also much larger under intraday checks (−2.72 log points) than under end-of-day checks (−0.97; Figure 15). The split

FIGURE 14

When intraday checks began in March 2020, every futures trade in a ban started to close positions

Monthly median of $R = (\Delta OI + V)/V$ on ban days, near and next futures, expiry days excluded

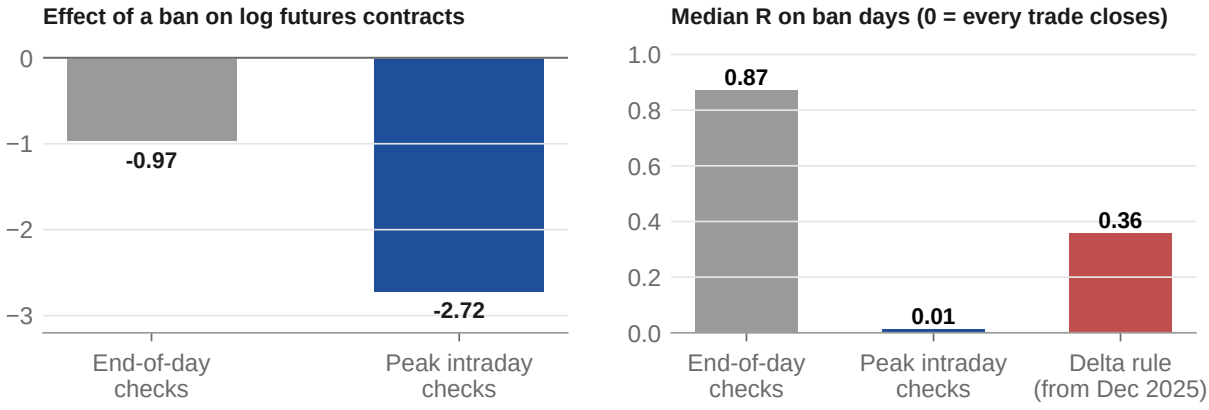


$R = 0$ when every trade closes a long and a short; $R \approx 1$ on normal days. Source: NSE F&O bhavcopies and ban lists; author's calculations.

FIGURE 15

The ban bites as designed only when positions are checked during the day

Effect of a ban on log futures contracts (left) and median R on ban days (right), by how violations were judged



End-of-day checks to 22 March 2020; peak intraday checks from 23 March 2020 to September 2025; delta rule from October 2025 (from 8 December 2025 shown). Source: author's calculations.

by enforcement period was not in the plan; I added it after the pooled result to find its cause. It shows that the rule changes trading as designed only when positions are checked during the day.

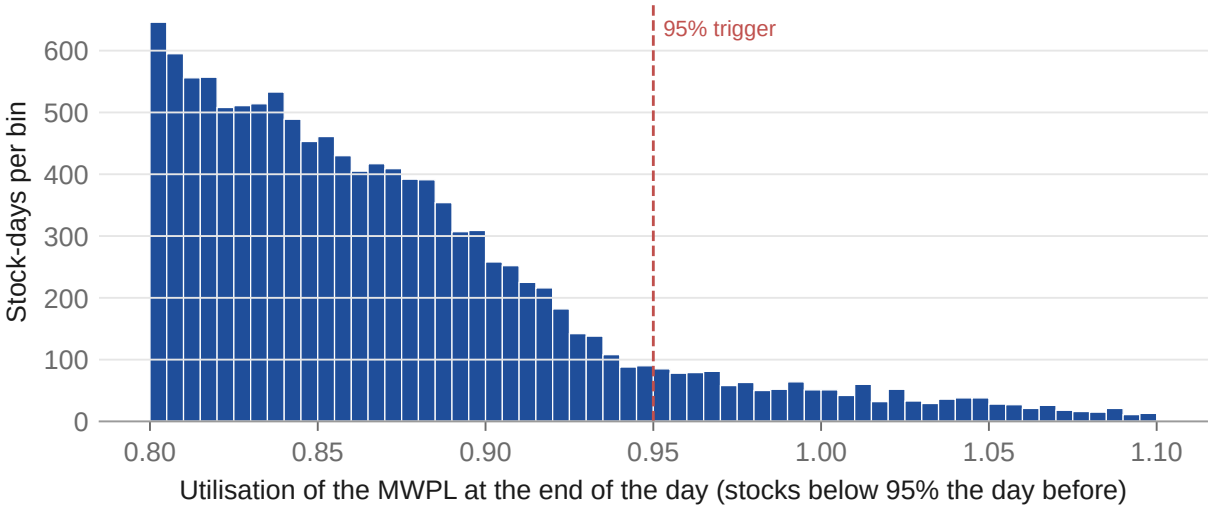
8.3 At the 95% trigger

Do traders hold back? If traders wanted to avoid a ban, utilisation would pile up just below 95% and thin out just above it. The model does not predict which way this goes (Section 4.4), so the test is descriptive. Its population needs care. Among all days on which a stock is not in

FIGURE 16

Utilisation does not bunch below the trigger

End-of-day utilisation of stocks that were below 95% the day before, January 2015 to September 2025



Cattaneo, Jansson and Ma test: $T = -0.18$, $p = 0.86$. Source: NSE combined open interest files; author's calculations.

a ban, utilisation above 95% is rare by construction, since a stock that crosses 95% is in the ban the next day. I therefore use the days on which a stock was below 95% the day before and ask how its utilisation is distributed today. The density is smooth through the trigger (Figure 16; the test of Cattaneo, Jansson and Ma (2020) gives $T = -0.18$, $p = 0.86$), and this holds across specifications and enforcement periods. Utilisation does not stall near the trigger; it speeds up. The share of stocks that cross 95% the next day rises steadily from about 7% at a utilisation of 85% to about one half at 93% to 95%. There is no sign that traders hold back to avoid a ban.

A sharp break in trading. The trigger can also be read as a regression discontinuity. For stocks not in a ban, ending a day above 95% puts the stock in the ban the next day in 99.7% of cases, against almost never just below. Comparing stocks just above and just below the trigger with local linear regressions, futures trading falls by 1.53 log points, R falls by 0.66 and the delivery share rises by 5.1 percentage points, while cash volume does not jump (Figure 17). These jumps confirm the results of the regression above with a different comparison: stocks that differ only in whether they ended the previous day just above or just below the trigger.

8.4 How the 2025 rules changed the number of bans

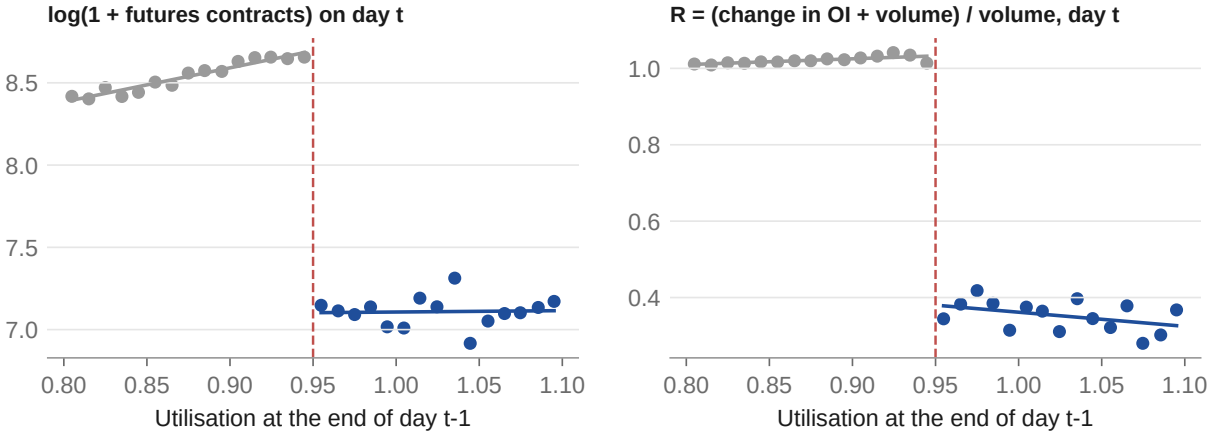
SEBI's stated aim in redesigning position limits in 2025 was to reduce "spurious" bans. Before the reform, bans came from rising open interest rather than from changes in the limit: on the day utilisation crossed 95%, the limit had fallen for only 8 of 1,744 entries, and only 7 of them (0.4%) would not have crossed at the previous day's limit. The reform changed two things at once: open interest came to be measured in delta-adjusted terms, and the new formula cut every stock's limit, by a quarter to a half (Figure 18).

To separate the two parts, I simulate the ban rule (enter the day after utilisation exceeds 95%,

FIGURE 17

Futures trading drops sharply at the trigger

Binned means by utilisation at the end of the previous day, stocks not in a ban the day before, expiry days excluded, with linear fits on each side

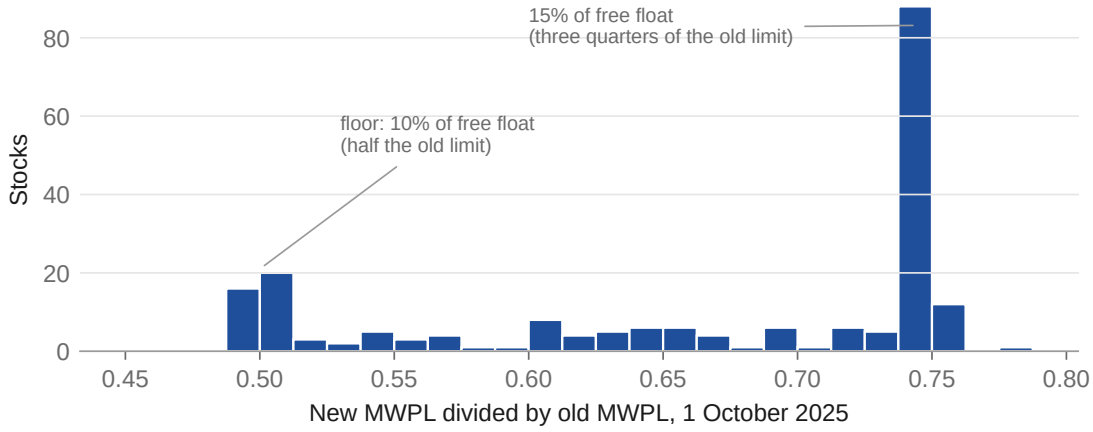


Estimates in the text are local linear with robust bias-corrected inference (Calonico, Cattaneo and Titiunik, 2014). January 2015 to September 2025. Source: author's calculations.

FIGURE 18

The October 2025 formula cut every stock's limit, by a quarter to a half

Ratio of the new to the old MWPL on 1 October 2025, all 208 F&O stocks



Source: NSE combined open interest files of 30 September and 1 October 2025; author's calculations.

leave the day after it is at or below 80%) for the stocks in the segment on both 30 September and 1 October 2025. Applied to the actual utilisation after the reform, the simulation reproduces the official ban lists exactly. I then apply it with open interest measured gross or in delta terms, and with the new limit or the limit the old formula would have given.[3] Under the old rule the same stocks would have spent about 3.7 times as many days in a ban in the first year (Table 10). The two parts of the reform pull in opposite directions. Measuring open interest in delta terms removes almost every ban, because options now count only by their delta and offsetting positions of the same client net out; the lower limit on its own would have quadrupled them. The combination cut

[3]The old-formula limit is approximated by scaling each stock's new limit by its ratio of old to new limit on 1 October 2025.

TABLE 10

The new rule produced about a quarter of the ban days the old rule would have

Bans from 1 October 2025 to 27 October 2026 for the stocks in the segment on 30 September and 1 October 2025

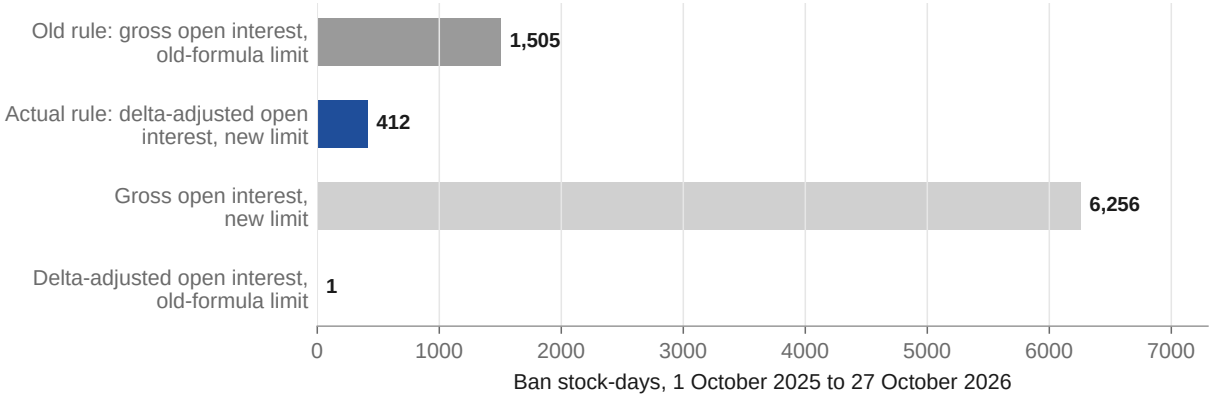
OPEN INTEREST AND LIMIT	BAN STOCK-DAYS	SPELLS	STOCKS
Delta-adjusted, new limit (the actual rule)	412	45	11
Gross, old-formula limit (the old rule)	1,505	112	36
Gross, new limit	6,256	296	85
Delta-adjusted, old-formula limit	1	1	1

Simulated ban rule, validated against the official lists. The 45 spells include one that began on 30 September 2025. Source: NSE combined open interest files; author's calculations.

FIGURE 19

Measuring open interest in delta terms removes almost all bans; the lower limit alone would have multiplied them

Ban stock-days, 1 October 2025 to 27 October 2026, under the actual and three counterfactual rules



Source: as for Table 10.

ban days by about three quarters (Figure 19). Whether the new rules have also changed what a ban does to prices needs more post-reform data, and is part of the second stage.

9. CONCLUSION AND THE SECOND STAGE

9.1 What this stage shows

This report set out to measure the band within which Indian stock futures are priced relative to their stocks, to ask whether costs and frictions set that band, and to ask whether the F&O ban restricts trading as intended. The answers are as follows.

The basis of Indian stock futures moves in a wide, asymmetric band, from about 35 bp below each stock's typical level to about 24 bp above it. The band is nonlinear in the way theory predicts, it is stable over time apart from the turmoil of 2020, and its lower side reaches further, as expected when the trade that corrects cheap futures needs a short sale. It is much wider than the intraday

bands found in earlier Indian studies, because an arbitrageur who carries a position overnight pays the taxes on delivery. Daily prices contain enough noise that the speed of reversion should be read with care; where the band lies is the more reliable finding.

Costs shape the band, but loosely. The full cost of a new cash-and-carry position lies in the outer tail of the basis, and the much smaller cost of rolling an existing position helps explain why the estimated edge lies inside it. The upper part of the basis rose after the 2024 and 2026 tax increases, as predicted and as the test fixed in advance confirms, but by more than the cost changes and by amounts that are not unusual on other dates. Across stocks, the band is wider where trading is harder.

The ban binds hard on trading. Futures trading falls by about 87% on ban days, cash trading moves toward delivery, and once violations were judged on intraday positions every futures trade during a ban closed an existing position. Enforcement therefore matters as much as the rule itself. Traders do not hold back before the trigger, and the 2025 rules cut the number of ban days by about three quarters, mainly by measuring open interest in delta terms.

9.2 Limitations

The data are daily, so trading within the day is outside the scope of this report, and noise in daily prices inflates measured reversion. The positions of arbitrageurs are not observed, so the model's prediction that the band moves with the existing book can only be checked indirectly. The tax increases are three events, one of which coincides with a change in fund taxation, so the test of costs has little power against shocks common to all stocks. The period after the 2025 reform is short.

9.3 The second stage

The first stage establishes the model, the test plan, the data, the band and the ban's effect on trading. The second stage uses these to ask what the limits do to prices:

- **The ban and the basis.** Whether correction of the basis becomes one-sided in a ban (Proposition 4, H7, a primary hypothesis), how the level of the basis changes when a ban starts and whether this depends on the composition of trading (Proposition 5), the 95% trigger as a discontinuity in prices, and placebo bans at false dates.
- **Position limits before a ban.** Whether the basis becomes richer as utilisation approaches the trigger, and what the 2025 formula and the 2020 halving and restoration of limits did to prices.
- **Arbitrage capital.** Whether flows into arbitrage funds narrow the band.
- **Short-sale constraints across the market.** Whether entry into the F&O segment, which brings the right to sell short, speeds the response of prices to news (H6, a primary hypothesis), whether anomalies are stronger where shorting is hardest, and a measure of short-sale constraints for every listed stock.
- **Policy.** What the results imply for the design and enforcement of position limits.

REFERENCES

Aggarwal, N. (2015). Limits to arbitrage: The case of single stock futures and spot prices. IGIDR Working Paper WP-2015-010.

Amihud, Y. (2002). Illiquidity and stock returns: Cross-section and time-series effects. *Journal of Financial Markets* 5(1), 31-56.

Calonico, S., M. Cattaneo and R. Titiunik (2014). Robust nonparametric confidence intervals for regression-discontinuity designs. *Econometrica* 82(6), 2295-2326.

Cattaneo, M., M. Jansson and X. Ma (2020). Simple local polynomial density estimators. *Journal of the American Statistical Association* 115(531), 1449-1455.

Corwin, S. and P. Schultz (2012). A simple way to estimate bid-ask spreads from daily high and low prices. *Journal of Finance* 67(2), 719-760.

De Long, J. B., A. Shleifer, L. Summers and R. Waldmann (1990). Noise trader risk in financial markets. *Journal of Political Economy* 98(4), 703-738.

Diamond, D. and R. Verrecchia (1987). Constraints on short-selling and asset price adjustment to private information. *Journal of Financial Economics* 18(2), 277-311.

Dwyer, G., P. Locke and W. Yu (1996). Index arbitrage and nonlinear dynamics between the S&P 500 futures and cash. *Review of Financial Studies* 9(1), 301-332.

Gromb, D. and D. Vayanos (2010). Limits of arbitrage. *Annual Review of Financial Economics* 2, 251-275.

Kolari, J. and S. Pynnönen (2010). Event study testing with cross-sectional correlation of abnormal returns. *Review of Financial Studies* 23(11), 3996-4025.

Miller, E. (1977). Risk, uncertainty, and divergence of opinion. *Journal of Finance* 32(4), 1151-1168.

Nandan, T., P. Agrawal and S. Bhargava (2014). Mispricing in CNX Nifty futures: An empirical investigation. *Asia-Pacific Journal of Management Research and Innovation* 10(4), 413-422.

Saurav, S., P. Srivastava and S. K. Agarwalla (2026). Decoupled: Disagreement, short-sale constraints, and law-of-one-price violations around earnings announcements. Working paper, SSRN.

Shankar, R. L., G. Sankar and K. Kiran Kumar (2019). Mispricing in single stock futures: Empirical examination of Indian markets. *Emerging Markets Finance and Trade* 55(7), 1619-1633.

Shleifer, A. and R. Vishny (1997). The limits of arbitrage. *Journal of Finance* 52(1), 35-55.

Slivka, R., V. Shah and J. Wu (2011). Arbitrage of single stocks versus futures in India. Working paper, Polytechnic Institute of New York University.

Ters, K. and J. Urban (2020). Estimating unknown arbitrage costs: Evidence from a 3-regime threshold vector error correction model. *Journal of Financial Markets* 47, 100503.

Tse, Y. (2001). Index arbitrage with heterogeneous investors: A smooth transition error-correction analysis. *Journal of Banking and Finance* 25(10), 1829-1855.

Regulatory documents. SEBI circular SEBI/HO/MRD/MRD-PoD-3/P/CIR/2024/1 (5 January 2024), short selling; SEBI/DNPD/Cir-31/2006 (22 September 2006), MWPL; SEBI press releases 18/2020 (20 March 2020) and 59/2020 (25 November 2020); SEBI circular SEBI/HO/MRD/TPD-1/P/CIR/2025/79 (29 May 2025), position limits and the ban rule; NSE Clearing circular 119/2025 (8 September 2025) and its FAQ (13 October 2025); NSE circular 1/2018 on STT for physically settled derivatives; NSE circulars FATAX56235 and FATAX73524 (STT from April 2023 and April 2026).

A. PROOFS

This appendix proves the propositions of Section 4. Notation is as in the text: market clearing gives $m = (u - a)/\Phi$ (2), and the arbitrage sector maximises (3).

Proposition 1. The objective (3) is strictly concave in a, with a kink at zero. Its best response to a basis m is $a = (m - c_+)/\gamma$ if $m > c_+$, $a = (m + c_-)/\gamma$ if $m < -c_-$, and $a = 0$ otherwise. In equilibrium $m = (u - a)/\Phi$. The map $a \mapsto \mathrm{BR}\big((u - a)/\Phi\big) - a$ is strictly decreasing, so the fixed point is unique. Substituting the clearing condition into each branch of the best response gives (4). For the upper branch, $m\Phi = u - (m - c_+)/\gamma$, so $m = (\gamma u + c_+)/(\gamma\Phi + 1)$, which equals c_+ at $u = \Phi c_+$, so the basis is continuous at the edge. The lower branch is symmetric. Inside the band $a = 0$ and $m = u/\Phi$.

Proposition 2. (i) and (ii) follow from (4): the edges, c_+ and $-c_-$, do not contain γ, and the slope outside the band, $\gamma/(\gamma\Phi + 1)$, is increasing in γ. (iii) The costs can be written as $c_+ = \tau_+ + (r_A - r)(T - t)$ and $c_- = \tau_- + (\ell_s + r - r_L)(T - t)$, where $\tau_\pm$ are the transaction costs of each trade, r_A is the return required on arbitrage capital, ℓ_s is the fee for borrowing the stock and r_L the interest earned on the proceeds of a short sale. Both edges are therefore linear in the time to expiry $T - t$ and converge to the transaction costs at expiry. (iv) The midpoint of $[-c_-, c_+]$ is $(c_+ - c_-)/2$, which is negative when $c_- > c_+$.

Proposition 3. With an existing position $a_{t-1} \geq 0$, the sector maximises $a_t m_t - \frac{\gamma}{2} a_t^2 - c_+(a_t - a_{t-1})^+ - c_u(a_{t-1} - a_t)^+$. This is strictly concave with a kink at a_{t-1}. The marginal value of holding one more unit is $m_t - \gamma a_t$, so adding pays only if $m_t - \gamma a_{t-1} > c_+$ and unwinding only if $m_t - \gamma a_{t-1} < -c_u$; otherwise the sector holds a_{t-1}. When it adds, the interior solution is $a_t = (m_t - c_+)/\gamma$; when it unwinds, $a_t = \max\{0, (m_t + c_u)/\gamma\}$. Combining each case with the clearing condition gives a unique equilibrium, as in Proposition 1.

Proposition 4. In a ban, restriction (B1) removes the option to add. Starting from $a_{t-1} > 0$, the sector holds its position as long as $m_t \geq \gamma a_{t-1} - c_u$. In this region $m_t = (u_t - a_{t-1})/\Phi$, so every increase in u_t passes through to the basis at the rate $1/\Phi$, instead of the smaller rate $\gamma/(\gamma\Phi + 1)$ that applies above the upper edge in normal times: rich futures are not corrected. If u_t is low enough that the sector unwinds, then $a_t = (m_t + c_u)/\gamma$ when interior, so $m_t = \gamma a_t - c_u$: cheap futures are corrected, at a floor that falls as the position shrinks. When a_t reaches zero, the floor disappears and $m_t = u_t/\Phi$. Positions that reach expiry during a ban are delivered and cannot be opened again in the next contract, so they leave the book for the rest of the ban.

Proposition 5. Split outside traders' demand shocks into closing demand ε^c and demand for new futures exposure $\varepsilon^n \geq 0$. In normal times both enter u one for one. In a ban, by (B2), new demand cannot be executed in futures; a share χ of it buys the stock instead, which in the units of (2) enters u as $-\lambda\varepsilon^n$ with $\lambda = \chi\psi_F/\psi_S \geq 0$. Compare the basis with and without a ban for the same starting position, the same initial pressure u^0 and the same shocks: $m^N = m(u^0 + \varepsilon^c + \varepsilon^n)$ and $m^B = m^{\text{ban}}(u^0 + \varepsilon^c - \lambda\varepsilon^n)$. Since m^N is non-decreasing and m^B non-increasing in ε^n, the

difference $m^B - m^N$ is non-increasing in ε^n. At $\varepsilon^n = 0$ the two inputs are equal, and since (B1) only removes the option to add, which lowers the basis when used, $m^B \geq m^N$. As ε^n grows, m^N rises without bound (its slope is at least $\gamma/(\gamma\Phi + 1) > 0$) while m^B does not rise, so $m^B < m^N$ for large enough ε^n. The sign of the ban's effect on the level therefore depends on how much of the pressure is new demand rather than closing demand.

Proposition 6. If every trader may only reduce positions in each contract, a buyer can only be closing a short and a seller only closing a long, so every trade lowers open interest by its size: $\Delta OI = -V$ outside expiry days, and $R = (\Delta OI + V)/V = 0$. Trades require that some holders of long positions and some holders of short positions both want to close at the same price, so volume falls. If an entity may instead lower its overall delta by opening an offsetting contract in another instrument, as allowed from October 2025, a trade can open a position in one contract, and the identity no longer holds.

B. THE TEST PLAN AND THE DATA

B.1 The pre-analysis plan

Before relating any outcome (the basis, trading, returns) to any treatment (ban status, utilisation, changes in the limit, tax dates), I wrote a plan that fixes the samples, the definitions of outcomes and treatments, the estimating equations, the predicted signs, the results that would contradict each prediction, and the method of inference. It was frozen with a time-stamped tag in the project's version control on 5 November 2026, after my supervisor's go-ahead and a full set of checks on the data. The plan has four primary hypotheses (H3, H6, H7 and H8), corrected jointly by Holm's method at 5%, and a set of secondary tests reported without adjustment. Standard errors are clustered by stock and by date unless stated otherwise, and the basis is winsorised at 1% and 99% within each year.

Where the plan left a detail open (for example, the bins of utilisation in the trading test, the windows and placebo dates of the cost test, or the population for the density test), I wrote the choice into a dated log and committed it before running the test. Analyses added after a result was seen, such as the split of the trading test by enforcement period and the noise correction of the threshold model, are identified as such in the text.

B.2 Data construction

Bans and position limits. The daily combined open-interest files give each stock's MWPL, its open interest, its futures-equivalent open interest from October 2025, and a flag for stocks in which no fresh positions may be opened the next day. Some files carry the year in two digits, a few carry another day's data under their own date, and some archives bundle several days; dates are therefore taken from the file names and checked. The official ban lists are missing at the exchange on three days; the lists for those days are imputed from the previous day's flags and continuing spells, and marked.

F&O membership. A stock is in the F&O segment on a day if its futures are listed that day. Symbol changes are linked through each security's identifier (ISIN), so that a renamed stock is not counted as an exit and an entry.

Prices and the basis. Prices, traded value and quantity, contracts and open interest come from the daily futures and cash bhavcopies, which changed format in July 2024. The older futures files do not report lot sizes; I recover them from open interest and check them against the day's price range. The futures volume-weighted price is traded value divided by contracts times lot size, and the cash price is traded value divided by quantity. Dividends going ex before expiry are subtracted at their present value. The interest rate is the monthly three-month interbank rate, carried forward for at most two months when the latest value is not yet published.

Reproducibility. Every data set is rebuilt from the raw files by scripts that report counts and checks, and every estimate in this report is produced by one sequence of analysis scripts. A complete rerun from the processed data reproduced every reported number.